

Does an LLM design better quantum circuits, or just bigger ones?

A five-gate, fixed-length comparison of LLM-guided and classical circuit search

Headline: last week's LLM advantage was mostly a circuit-size decision. Remove that one degree of freedom and it disappears on three of four tasks.

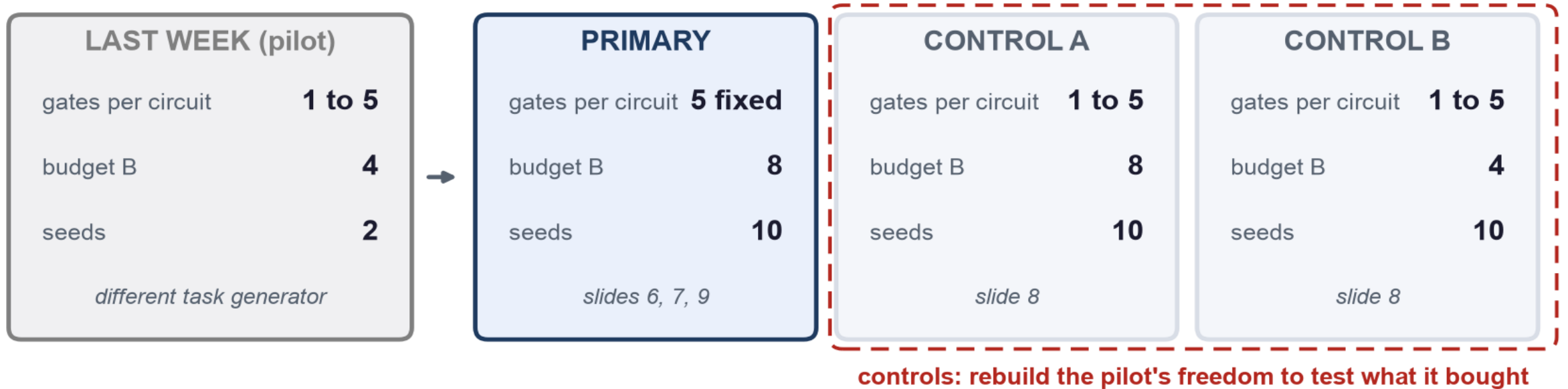
Tomoya Hatanaka

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WHAT CHANGED

Last week the LLM could choose how big the circuit was



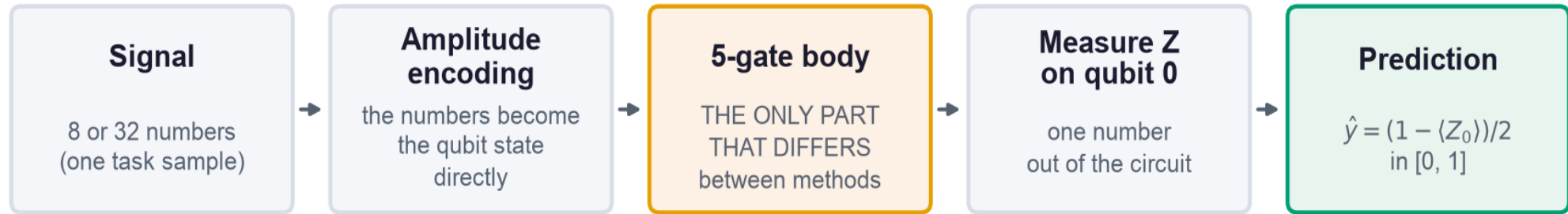
Longer circuits usually score better. So a proposer that always asks for the maximum length wins without designing anything. Fixing the length takes that move away.

Every rule, stated so you can check any circuit by eye

Constraint	Value
Tasks	T1 peak position · T2 sinusoid frequency · T3 change-point location · T4 one peak vs two (binary)
Qubits	3 for T1, T2 · 5 for T3, T4 (a circuit on n qubits holds 2^n numbers)
Gate count	5 exactly (PRIMARY) · 1 to 5 (both controls)
Gate set	H, RX, RY, RZ (one qubit) · CRX, CRY, CRZ (two qubits)
Angles	one number per gate, anywhere in [-pi, pi] (H has none)
Budget B	8 distinct circuits scored per method (4 in Control B) · 10 seeds
Training	none - proposed angles are used exactly as written
Encoding / readout	amplitude encoding; measure Z on qubit 0
Classical params	zero - nothing outside the circuit can fix a bad circuit

One operation = one physical gate; a 5-gate circuit compiles to 5 gates. The same grammar check is applied to classical samplers and LLM replies alike

One number: RMSE, the typical size of the error



No training anywhere: the angles a method proposes are used exactly as written. No classical layers: nothing outside the circuit can fix a bad circuit.

Validation RMSE

drives the search - each method keeps its own best

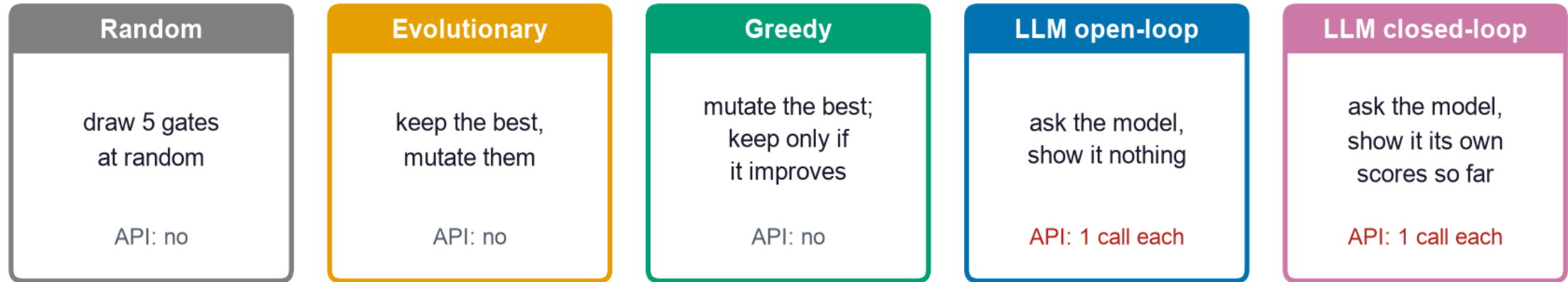
Test RMSE

reported once, after the choice is frozen. Never used to choose.

T4 is yes/no

so RMSE there is the error on a 0/1 label. AUROC is shown too - slide 7.

Five methods, one shared budget



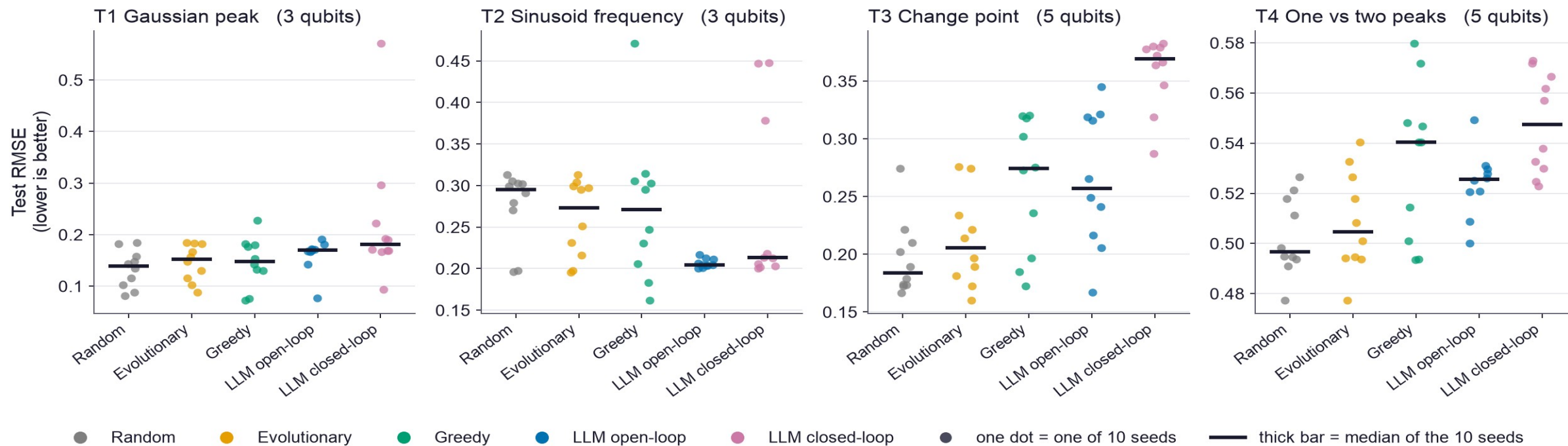
All five get the same budget B: the same number of DISTINCT circuits actually scored. Repeats and malformed proposals are free but earn nothing.

Open-loop vs closed-loop: open-loop never sees how its own circuits scored; closed-loop does, eight times. That is the only difference between them.

RESULT

PRIMARY: exactly 5 gates, B = 8, 10 seeds all four tasks, T1-T4

Random search wins on three of four tasks

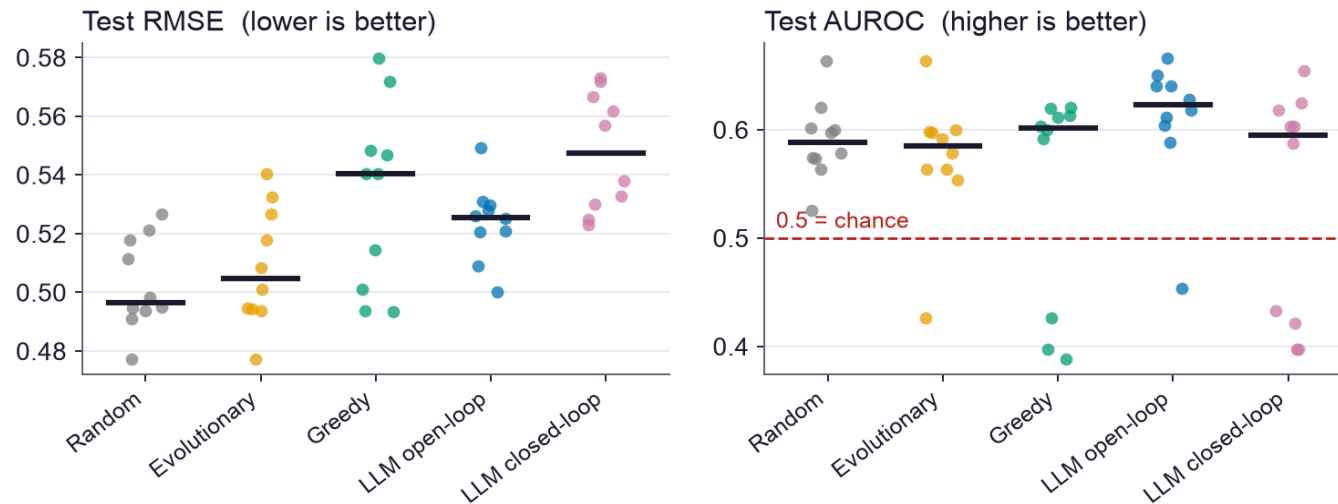


Each panel has its own y-axis: RMSE is comparable between arms within a panel, never between panels, because the tasks regress different quantities.

The LLM wins on T2 alone, and there it wins clearly (0.204 vs 0.295). Closed-loop never beats open-loop on any task: eight rounds of feedback did not help this model at this budget.

On T4 the metric, not the method, picks the winner

T4 is binary classification: RMSE against a 0/1 label is $\sqrt{\text{Brier}}$, a proper score, but it mixes calibration with discrimination.



AUROC

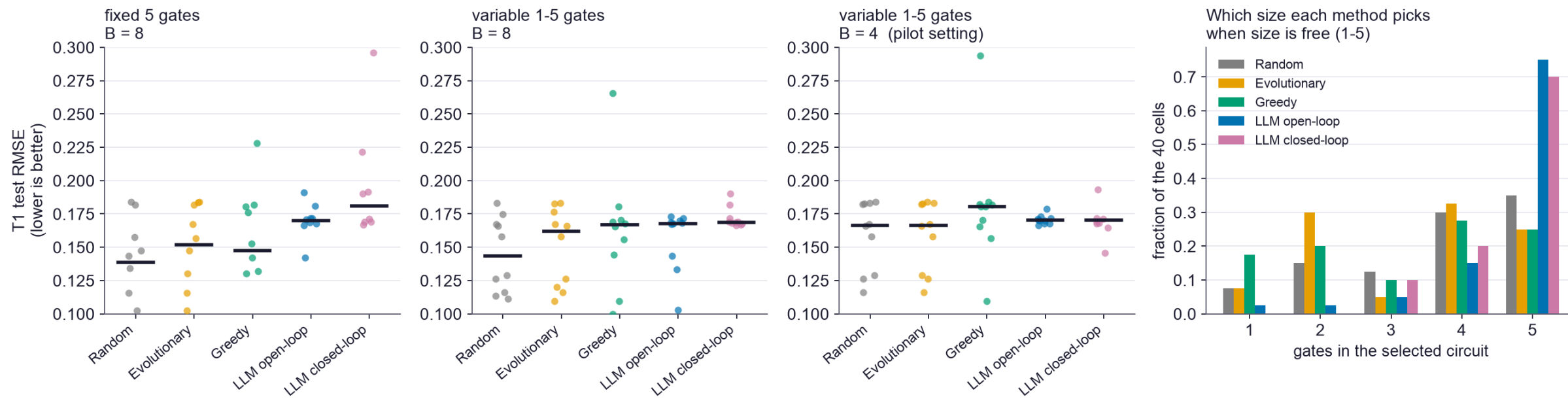
= how often the model ranks a two-peak signal above a one-peak one. 1.0 perfect, 0.5 coin flip.

Why they disagree

RMSE also punishes outputting 0.45 / 0.55 instead of 0 / 1. AUROC only cares about the order.

Neither reading is strong: AUROC spans 0.59 to 0.62 against 0.5 for chance. Five gates without training barely solve T4 at all.

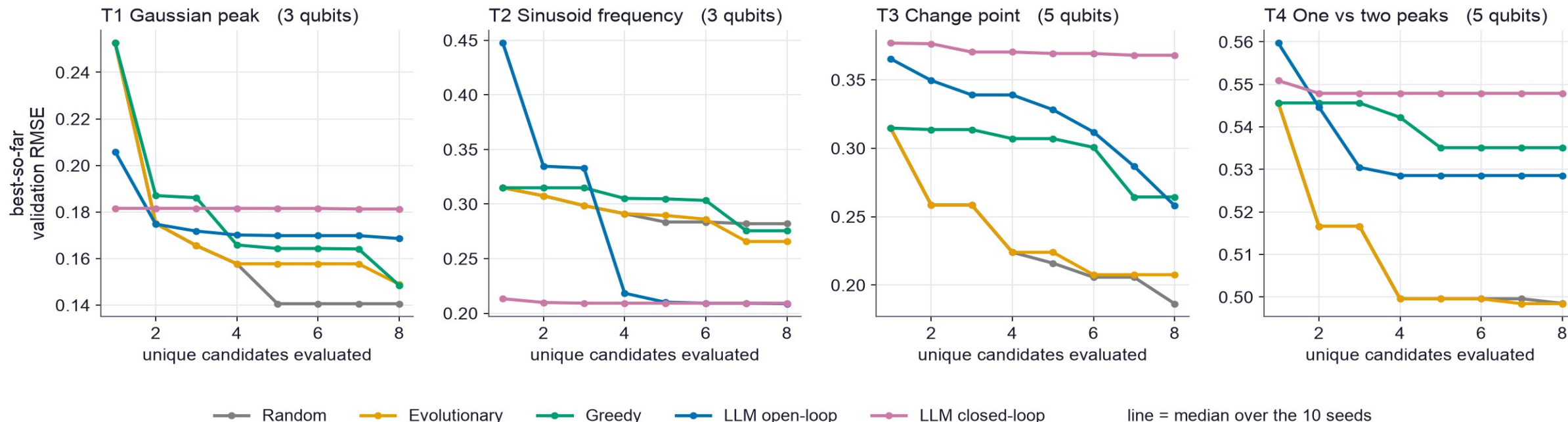
Last week's advantage was a size decision, not a design one



Left three panels: one dot = one of 10 seeds, thick bar = median. Only Random degrades as circuit size becomes a free choice, because the LLM almost always picks the maximum size while Random draws it uniformly. The pilot rewarded that single decision.

Free the length and only Random gets worse (0.139 → 0.167); the LLM does not move (0.170 → 0.170), because it was already asking for 5. At the pilot's setting they meet.

More feedback did not translate into better circuits



Eight rounds of feedback, no gain on any task. A first run had closed-loop repeating itself 74% of the time and starved of budget; the prompt now lists what it already tried, repeats fell to 16%, and these are the corrected numbers.

What I can and cannot claim

Supported

- Length fixed: no win over random on 3 of 4 tasks
- Clear win on T2, across all 10 seeds
- Length free: the LLM picks the maximum ~70% of the time
- That one choice explains most of last week's advantage

NOT supported

- "LLMs cannot design circuits" - one small model, one budget
- "Feedback does not work" - it failed here, at this scale
- Anything about hardware - this is exact simulation
- Comparing RMSE across different tasks

Next: give the LLM a decision that is actually hard, and check any advantage the same way.